

Chapter 10 - TVTP Implementation Note

Macroeconometrics companion code | Python, R and MATLAB

Purpose. This note documents a correction to the time-varying transition probability (TVTP) mapping in the Chapter 10 Markov-switching Gibbs sampler. It also records the cross-language consistency check performed on the Python, R and MATLAB companion implementations.

1. Issue identified

The Hamilton filter used in the Chapter 10 code expects $p = P(s_t=0 | s_{t-1}=0)$ and $q = P(s_t=1 | s_{t-1}=1)$. In the original Python and R TVTP blocks, the same logistic probability was assigned to both p and q . This is inconsistent with the transition matrix used by the filter.

For the specification

$$P(s_t=1 | s_{t-1}) = \text{sigma}(g_0 + g_1 s_{t-1}),$$

the corresponding stay probabilities are

$$p_{00} = 1 - \text{sigma}(g_0) \text{ and } q_{11} = \text{sigma}(g_0 + g_1).$$

2. Correction applied

Python: the TVTP block now constructs p_vec from $1 - \text{sigma}(g_0)$ and q_vec from $\text{sigma}(g_0 + g_1)$. No other econometric specification, data transformation, prior, lag order, MCMC setting, regime-specific VAR equation or IRF calculation was intentionally changed.

R: the corresponding TVTP block has been changed to the same p_{00}/q_{11} mapping. The previous line assigning q_vec equal to p_vec has been removed.

MATLAB: no mathematical correction was required. The `msvar` toolkit already constructs distinct transition probabilities and passes p and q separately to the Hamilton filter. The revised MATLAB main script therefore adds a documentation comment recording this verification, while retaining the existing estimator.

Implementation	Status	Action
Python	Correction required	Use $p_{00} = 1 - \text{sigma}(g_0)$; $q_{11} = \text{sigma}(g_0 + g_1)$
R	Correction required	Use the same p_{00}/q_{11} mapping as Python
MATLAB	Already distinct	No estimator change; verification note added

3. Consequence for numerical results

This is a substantive implementation correction rather than a portability or software-version adjustment. Accordingly, corrected Python and R MS-VAR estimates can differ from results produced by the original Python/R companion code. The corrected results should be treated as the current companion-code results.

An external reproducibility check reported that the corrected mapping produced regime persistences close to independent maximum-likelihood implementations and substantially above the approximately 0.5 values generated by the mismatched mapping. That external validation is useful supporting evidence, but it is reported here as an external check rather than as a numerical result independently reproduced in preparing this note.

4. Suggested repository note

Chapter 10 - MS-VAR TVTP correction. The Python and R implementations of the time-varying transition probabilities have been corrected so that the Hamilton filter receives $p = P(s_t=0 \mid s_{t-1}=0)$ and $q = P(s_t=1 \mid s_{t-1}=1)$. For $P(s_t=1 \mid s_{t-1}) = \sigma(g_0 + g_1 s_{t-1})$, the corrected mapping is $p = 1 - \sigma(g_0)$ and $q = \sigma(g_0 + g_1)$. This correction can change the numerical MS-VAR results. The MATLAB msvar implementation already distinguishes the two persistence probabilities and requires no corresponding estimator change.

5. Reproducibility recommendation

For the final companion package, retain the corrected Python and R files together with the verified MATLAB file and this note. When reporting updated numerical results, use identical data, sample, lag order and MCMC settings across implementations, and distinguish Monte Carlo variation from implementation differences.